

Karthik Bharadwaj

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EDUCATION

Northeastern University, Boston, MA

Expected May 2027

Master of Science in Computer Science | **GPA: 4.0**

Coursework: Algorithms, Database Management Systems, Foundations of AI, Programming Design Paradigms

R V College of Engineering, Bangalore, India

2023

Bachelor of Engineering in Computer Science and Engineering

EXPERIENCE

Digital Associate – Software Engineer

Sept. 2024 – June 2025

Digitide Solutions Ltd (Client: AB-InBev)

Bangalore, India

- Engineered a modular SaaS platform serving **1,000+ daily users** with report generation, admin controls, planogram extraction, and bulk update/delete workflows using Angular, Python, Docker, and Azure.
- Developed an AI-powered chatbot using NLP and Generative AI, enabling warehouse staff to query inventory and planogram data via natural language—reducing manual lookup time by **60%** and eliminating engineering intervention on routine data queries.
- Designed and integrated real-time operational dashboards consuming sensor and transactional data streams, enabling proactive alerting and data-driven decision-making across warehouse operations.

Software Engineer – Full Stack

Jan. 2024 – July 2024

Zevvo

Bangalore, India

- Architected core backend systems end-to-end—bookings, billing, subscriptions, and catalog—with MongoDB schema design and JWT authentication resistant to XSS/CSRF; onboarded **50+ active users** within the first month and reduced auth-related bugs by **40%**, improving platform reliability and user trust.
- Built a full admin portal and customer-facing pages from scratch using React, Node.js, and Express, delivering the product from zero to production in under **6 months**.
- Automated CI/CD pipelines via GitHub Actions deploying Dockerized services to AWS ECS with auto-scaling, increasing deployment frequency by **40%** and enabling same-day hotfixes for production issues.

Software Development Intern

March 2023 – July 2023

Crestron Electronics

Bangalore, India

- Developed and deployed .NET Core REST APIs with optimized SQL queries, cutting average response time by **28%**, directly improving page load speeds for end users across the platform.
- Built mobile-responsive React components with Google Tag Manager integration, driving behavioral analytics that informed the product feature roadmap.
- Collaborated with cross-functional teams to integrate analytics tracking across **3 customer-facing modules**, providing product managers with actionable usage data that shaped the feature roadmap.

PROJECTS

ParetoFolio – Portfolio Optimization Engine ([Link](#))

Fall 2025

Tech Stack: TypeScript, React 19, FastAPI, Python, Pydantic v2

- Built a responsive React 19 frontend with interactive data visualization and a typed FastAPI backend, serving a RESTful API for multi-objective portfolio optimization across **75 US equities**.
- Implemented NSGA-II from scratch—SBX crossover, polynomial mutation, tournament selection, and crowding distance—producing Pareto-optimal portfolios with an interactive efficient frontier and Sharpe-ranked table.

SportsTV Streaming Data Warehouse ([Link](#))

Fall 2025

Tech Stack: R, MySQL 8, SQLite, R Markdown, Kimball Star Schema

- Engineered a bulk-batched, idempotent ETL pipeline with surrogate-key resolution and partition pruning, achieving **99%+ match rate** across **2M+ streaming transactions** from heterogeneous sources.
- Modeled a Kimball-style star schema with five conformed dimensions and a RANGE-partitioned fact table; generated a fully automated BI report driven entirely by live SQL queries.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, C#

Backend: Node.js, .NET Core, Express.js, FastAPI, REST APIs, Microservices

Frontend: React.js, Next.js, Angular, React Native, HTML/CSS, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, Firebase, MSSQL

Cloud / DevOps: AWS (ECS, S3, Lambda, EC2, ECR), Azure (Container Registry, Blob Storage), GCP, Docker, Kubernetes, GitHub Actions, CI/CD

Practices: System Design (HLD/LLD), Agile/Scrum, TDD, OOP, Code Reviews, Git